The University of Bristol

Mumbai Enterprise Campus

Faculty of Science and Engineering
School of Engineering Mathematics and Technology
SEMTM0056: Introduction to Financial Technology (INFT)
Academic Year 2026-27

Main Coursework

Release date: Friday 23/10/2026.

Submission: Q1: 13:00 Wed 28/10/2026; Q2: 13:00 Tue 03/11/2026; Q3: 13:00 Fri 06/11/2026.

- **What to submit:** PDF report and all code used to generate your results.
- **Maximum page lengths**: **Q1:** 1-page, **Q2**: 2-pages, **Q3**: 2-pages

Marks available: 100

Individual assessment: You should work alone on this assessment.

PDF report: Submit your answers to each question as a short report in a single PDF file, **following the maximum page limits given** above. Your report should use 11-point Arial font for the main body text and margins of 2 cm on all sides. Do not reduce the main body font size or margins or otherwise alter the template to circumvent the page limit. Such formatting may result in part of the report not being considered for assessment. Reasonably smaller text may be used in figures, captions and compact tables, provided it remains clearly legible. The report, including all figures and tables, must not exceed the page limits. **Material beyond the page limit will not be considered for assessment**.

Code files: Submit the code used to generate your reported results, including clearly named Jupyter notebooks, your BSE.py file, any other Python files you write, and any input price data used. The submitted code is supporting evidence and will be used where necessary to check the correctness and reproducibility of your work. It is not necessary to produce polished or extensively documented notebooks, and notebooks are not assessed as additional report content. All material that you wish to be considered for marks must be presented in the 5-page report.

Experimental validity: You are responsible for ensuring that your experiments generate sufficient market activity and independent evidence to support the conclusions you draw. A large number of repeated trials or a statistically significant result does not compensate for an experimental design that does not adequately test the claim being made.

Use of AI: Minimal.

Reproducibility and experimental method: Your report must provide sufficient methodological detail for another researcher to understand and reproduce the experiments supporting your conclusions. Clearly specify the experimental configurations, market populations, simulation duration, number of independent trials, parameter settings, data sources, performance measures, and analysis methods used. Your submitted code must reproduce the results presented in the report.

You should report your method concisely; detailed method does not require a chronological description of every experiment performed.

Question 1: BSE fundamentals and interpretation [30 marks]

You want to address the claim, “ZIP traders generate more profit than ZIC traders in homogeneous and periodic BSE markets with static and symmetric demand and supply curves”.

Using BSE, configure and run a series of experiments to address this claim. Present appropriate quantitative and statistical analysis and one figure that best summarises your experimental results.

Interpret and explain your findings. Your explanation should go beyond reporting statistical significance: use your understanding of ZIP and ZIC trader behaviour and the experimental market configuration to explain the behaviour you observe. You may use a maximum of one additional figure as part of your explanation. Additional figures will not be considered for assessment.

Question 2: Investigating and improving MMM01 [40 marks]

BSE includes a Minimal Market Maker MMM01, which has three configuration parameters with default settings: n_past_trades=1, bid_percent=0.5, ask_delta=25.

Conduct an empirical investigation of MMM01 and identify a parameter configuration that you believe represents a robust improvement over the default. You are responsible for designing an experimental programme capable of supporting this conclusion.

Your investigation must include both synthetic market conditions and at least one market that incorporates real pricing data as an input to offset supply and demand. Beyond these requirements, you should decide what experimental conditions and evidence are necessary to support your claims.

Explain and justify your experimental design, analyse the resulting market behaviour and profitability, and critically assess how robust your selected configuration is. Clearly state and justify the three parameter values selected for your robust configuration; this will be called MMM01*.

Important: Statistical significance alone is not sufficient evidence that a configuration is robust or superior.

Question 3: Developing and evaluating MMM02 [30 marks]

BSE also includes a Minimal Market Maker MMM02, initially containing logic equivalent to MMM01. Using what you learned from your investigation in Question 2, identify a limitation of MMM01* and modify the logic of MMM02 to address it.

Describe the new MMM02 logic and explain the rationale for your design. Your modification should be motivated by evidence and reasoning about market behaviour, rather than solely by an attempt to increase measured profit.

Design and conduct experiments to determine whether MMM02 represents a genuine improvement over MMM01*. Your evaluation should consider whether any apparent improvement generalises beyond the conditions in which it was developed.

Robustness challenge: Identify at least one plausible feature of your experimental design or market environment that could make MMM02 appear better than it really is. Design and run an additional experiment to test this possibility. Use the resulting evidence to explain both what your experiments demonstrate and what they do not demonstrate.

A successful modification is not required to receive credit. Marks are awarded for the quality of the rationale, implementation, experimental evaluation and conclusions supported by the evidence.

Assessment rubric

The criteria below show how marks are allocated. High marks require not only technically correct work, but conclusions that are supported by an appropriate experimental design and convincing evidence. Statistical significance alone does not demonstrate that a result is robust.

Q1: BSE fundamentals and interpretation [30 marks]

Criterion	Marks
Correct experimental design and BSE configuration	**10**
Quantitative/statistical analysis and presentation of evidence	**6**
Interpretation of experimental results	**4**
Explanation of observed market behaviour	**10**
Total	**30**

Important: Your experimental configuration must correctly address the stated claim. Fundamental configuration errors that result in a different market being investigated will substantially limit the marks available for analysis, interpretation and explanation.

Q2: Investigating and improving MMM01 [40 marks]

Criterion	Marks
Experimental design and reproducibility	**10**
Quality and adequacy of experimental evidence	**10**
Analysis and interpretation of results	**8**
Robustness of investigation and conclusions	**8**
Selection and justification of MMM01*	**4**
Total	**40**

Q3: Developing and evaluating MMM02 [30 marks]

Criterion	Marks
Rationale and design of MMM02	**7**
Implementation	**5**
Experimental evaluation	**6**
Robustness challenge	**6**
Interpretation, limitations, and conclusions	**6**
Total	**30**